

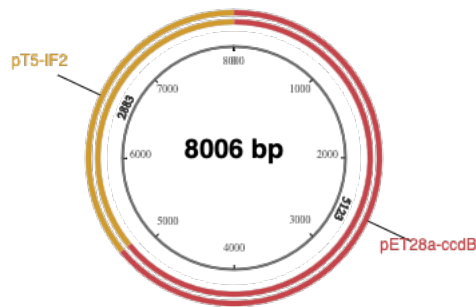
pET28a-pT5-IF2

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Component Fragments

Name	Length	Produced by	5' End	3' End
pET28a-ccdB	5143	PCR	Fwd Primer (auto)	Rev Primer (auto)
pT5-IF2	2903	PCR	Fwd Primer (auto)	Rev Primer (auto)



Notes

- Everything looks OK. No major issues detected.

Required oligos

Name	Primer 5' (overlap/spacer/ANNEAL) 3'	Len	%GC	3' %GC	3' Tm	3' Ta
pET28a-ccdB_fwd	tgtgcacctgGCTTGATCCGGCTGCTAAC	29	59	58	66.4	65.6
pET28a-ccdB_rev	atgaggacatATTTCGCGGGATCGAGATC	29	48	53	64.6	65.6
pT5-IF2_fwd	cccgcgaaatATGTCCTCATAGTTTGAACAAG	32	44	36	59.5	60.5
pT5-IF2_rev	cggatcaagcCAGGTGCACATTAAGCAATG	30	50	45	61.8	60.5

Build Settings

Property	Value
Product/Kit	#E5520 NEBuilder HiFi DNA Assembly Cloning Kit
Minimum Overlap	20 nt
Minimum Overlap Tm	48 °C
Circularize	Yes
PCR Polymerase/Kit	Q5 High-Fidelity DNA Polymerase
PCR Primer Conc.	500 nM
Min. Primer Length	18 nt

Assembled Sequence

```
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#ACCESSION   .
#VERSION     .
#KEYWORDS    NEBuilder
#SOURCE      synthetic DNA construct
# ORGANISM   synthetic DNA construct
#REFERENCE   1 (bases 1 to 8006)
# AUTHORS    .
# TITLE      NEBuilder-generated Construct
# JOURNAL     Exported 11-APR-2025 from NEBuilder https://nebuilder.neb.com
#COMMENT     NEBuilder-generated oligos (UPPERCASE = gene-specific, lowercase = overlap)
#COMMENT     pET28a-ccdB_fwd: tgtgcacctgGCTTGATCCGGCTGCTAAC
#COMMENT     pET28a-ccdB_fwd 3'Tm: 66.4 3'Ta: 65.6
#COMMENT     pET28a-ccdB_rev: atgaggacatATTTGCGGGGATCGAGATC
#COMMENT     pET28a-ccdB_rev 3'Tm: 64.6 3'Ta: 65.6
#COMMENT     pT5-IF2_fwd: cccgcgaaatATGTCCTCATAGTTTGAACAAG
#COMMENT     pT5-IF2_fwd 3'Tm: 59.5 3'Ta: 60.5
#COMMENT     pT5-IF2_rev: cggatcaagcCAGGTGCACATTAAGCAATG
#COMMENT     pT5-IF2_rev 3'Tm: 61.8 3'Ta: 60.5
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